25FEBS201ESE

23 APR 2026

Seat No. : 25301

Student Name : (real)

St. John College of Engineering and Management, Palghar, Maharashtra
Autonomous Institute

End Semester Examinations, April 2026

ASH - SEMESTER II

25FEBS201 - Engineering Mathematics II

Time : 2.00 Hours **Maximum Marks : 60**

Instructions

- All Questions are compulsory
- Assume suitable data wherever required

		Attempt all Questions	Mrk	Course Outcome	RBT Level
1	a	i. Apply **De Moivre's Theorem** to Expand $cos^6\theta$ in series of cosines of multiples of θ.	5	CO1	L3
		ii. Apply **De Moivre's Theorem** to solve $z^5 = 1$.	5	CO1	L3
		OR			
	b	Apply **De Moivre's Theorem** to Show that the roots of $(x+1)^7 = (x-1)^7$ are given by $-i\ cot\left(\frac{k\pi}{7}\right);\ k = 0, 1, 2, \ldots, 6.$	10	CO1	L3
2	a	i. In a circuit containing inductance L, Resisitence R and Voltage E, the current i is given by $L\frac{di}{dt} + R\,i = E$. Find the current i at time t if at $t = 0,\ i = 0$. Also find i when $L = 640\ henries, r = 250\ ohms\ and\ E = 500\ volts$	5	CO2	L3
		ii. Apply the appropriate definition and necessary conditions to solve the given differential equation $(y + x\ sin\ x)dx + (x - 2e^y)dy = 0$.	5	CO2	L3
		OR			
	b	Solve the linear differential equation $x\ cos\ x\ \frac{dy}{dx} + y\ (x\ sin\ x + cos\ x) = 1$	10	CO2	L3
3	a	i. Solve $(D^3 - 4D)y = e^{2x} + e^{-2x} + 2$.	5	CO3	L3
		ii. Solve $(D^2 - 2D + 2)y = x + e^x cos(x)$	5	CO3	L3
		OR			
	b	Solve $\frac{d^2y}{dx^2} + 2y = x^2e^{3x} + e^x - cos\ 2x$	10	CO3	L3
4	a	i. Evaluate $\int_0^1 \int_{x^2}^{x} (x^2y + xy^2)dydx$	5	CO4	L3
		ii. Evaluate $\int_1^2 \int_0^x \frac{dydx}{x^2+y^2}$	5	CO4	L3
		OR			
	b	Change the order of integration and evaluate $\int_0^5 \int_{2-x}^{2+x} dydx$	10	CO4	L3
5	a	i. Evaluate $\int_0^2 \int_0^z \int_0^{yz} xyzdxdydz$	5	CO5	L3
		ii. Evaluate $\int_0^1 \int_0^2 \int_1^2 x^2yzdzdydx$	5	CO5	L3
		OR			
	b	Evaluate $\int\int\int xyzdxdydz$ over the positive octant of the sphere $x^2 + y^2 + z^2 = 1$	10	CO5	L3
6	a	i. Solve the following using Modified Eulers Method, $\frac{dy}{dx} = x - y^2, y(0) = 1$ find y for $x = 0.05$.	5	CO6	L3

		Correct to four decimal places.			
		ii. Solve the following using Eulers Method, $\frac{dy}{dx} = x + y, y(0) = 1$ find y at $x = 1$ in 5 steps.	5	CO6	L3
		OR			
	b	Solve the following integral using a) Trapezoidal b) Simpson's 1/3 rd and c) Simpson's 3/8 th rule: $\int_0^1 \frac{x^2}{1+x^3} dx$	10	CO6	L3
* All the Best *					

